Problem 1.

a) $m\frac{d^2x}{dt^2} + c\frac{dx}{dt} + kx = A\sin(\omega t)$

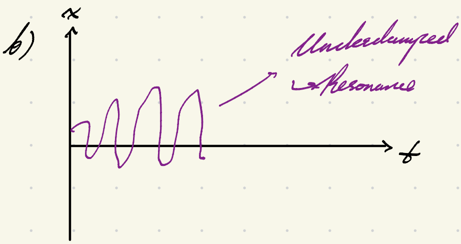
$\Rightarrow x'' + 0.2x' + x = \sin(\omega t)$

$u = x(t)$ → Position given time

$v = x'(t) = u'$ → Speed given time

$v' = x''(t) = A\sin(\omega t) - cv - ku$ → Acceleration given time

b)



$\zeta = \frac{c}{2\sqrt{mk}} = \frac{0.2}{2} = 0.1 < 1$

↳ Underdamped

→ Damping Reduces Amplitude

→ $\omega \approx \omega_0$

↳ External Force is adding energy

→ Constructive interference

Steady-state will still occur

↳ Damping enforces a boundary. Amplitude limited by damper.

c) $y_{n+1} = y_n + \frac{1}{6}(k_1 + 2k_2 + 3k_3 + k_4)$

	for u	for v	for x_{n+1}
k_1	$\Delta t \cdot v_n$	$\Delta t(\sin(\omega t_n) - 0.2v_n - u_n)$	
k_2	$\Delta t \cdot (v_n + \frac{k_{1v}}{2})$	$\Delta t(\sin(\omega(t_n + \frac{\Delta t}{2})) - 0.2(v_n + \frac{k_{1v}}{2}) - (u_n + \frac{k_{1u}}{2}))$	
k_3	$\Delta t \cdot (v_n + \frac{k_{2v}}{2})$	$\Delta t(\sin(\omega(t_n + \frac{\Delta t}{2})) - 0.2(v_n + \frac{k_{2v}}{2}) - (u_n + \frac{k_{2u}}{2}))$	
k_4	$\Delta t(v_n + \frac{k_{3v}}{2})$	$\Delta t(\sin(\omega(t_n + h)) - 0.2(v_n + k_{3v}) - (u_n + k_{3u}))$	

d) Case 1. $\Delta t = 1$

Case 2. $\Delta t = 0.1$

$\omega = 1$

$x(0) = 0$, $x_0 = 0$

$V(0) = 0$, $V_0 = 0$

Find x_1

$\rightarrow$ Coarse

$\rightarrow$

$$k_{1u} = \Delta t(V_0) = 0 \qquad k_{1V} = 0$$

$$k_{2u} = \Delta t\left(V_0 + \frac{k_{1u}}{2}\right) = 0 \qquad k_{2V} = \Delta t\left(\sin\left(t_0 + \frac{\Delta t}{2}\right) - 0.2\left(V_0 + \frac{k_{1V}}{2}\right) - u_0 + \frac{k_{1u}}{2}\right)$$

$$= \sin(0.5)$$

$$= 0.479$$

$$k_{3u} = \Delta t\left(V_0 + \frac{k_{2V}}{2}\right) = 0.240 \qquad k_{3V} = \Delta t\left(\sin\left(t_0 + \frac{\Delta t}{2}\right) - 0.2\left(V_0 + \frac{k_{2V}}{2}\right) - \left(u_0 + \frac{k_{2u}}{2}\right)\right)$$

$$= 1\left(\sin(0.5) - 0.2\left(\frac{0.479}{2}\right) - 0\right)$$

$$= 0.431$$

$$k_{4u} = \Delta t(V_0 + k_{3V}) = 0.431 \qquad k_{4V} = \Delta t\left(\sin(t_0 + \Delta t) - 0.2(V_0 + k_{3V}) - (u_0 + k_{3u})\right)$$

$$= 1(\sin(1) - 0.2(0.431) - 0.240)$$

$$0.515$$

$$u_1 = u_0 + \frac{1}{6}\left(k_{1u} + 2k_{2u} + 2k_{3u} + k_{4u}\right)$$

$$= 0.152$$

$$V_1 = V_0 + \frac{1}{6}\left(k_{1V} + 2k_{2V} + 2k_{3V} + k_{4V}\right)$$

$$= 0.389$$

$\rightarrow$ Fine

$\longrightarrow \Delta t = 0.1$

$\rightarrow$

$$k_{1u} = 0.1(V_0) = 0 \qquad k_{1V} = 0.1(\sin 0 - 0 - 0) = 0$$

$$k_{2u} = 0.1\left(V_0 + \frac{k_{1u}}{2}\right) = 0 \qquad k_{2V} = 0.1(\sin(0.05) - 0.2(0) - (0)) = 0.00500$$

$$k_{3u} = 0.1\left(V_0 + \frac{k_{2V}}{2}\right) = 0.000250 \qquad k_{3V} = 0.1\left(\sin(0.05) - 0.2\left(\frac{0.00500}{2}\right) - (0 + 0)\right) = .00495$$

$$k_{4u} = 0.1(V_0 + k_{3V}) = 0.000495 \qquad k_{4V} = 0.1(\sin(0.1) - 0.2(0 + 0.00495 - 0 + 0.000250)$$

$$= 0.000990$$

$u_1 = 1.66 \times 10^{-4}$

$V_1 = 0.00496$

e) ω : if system is near resonance, ω can force the system to constructively add or subtract.

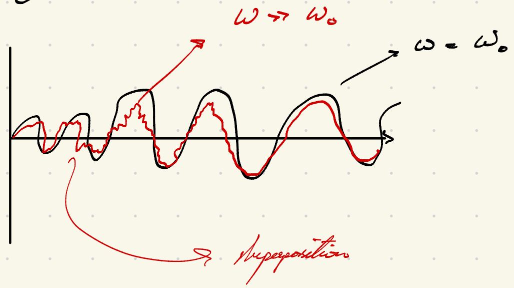
→ if ω is uncertain, system behaviour could be vastly different.

$\omega_0 = \sqrt{\frac{k}{m}}$; if $\omega \neq \omega_0$, system is very different.

f) Initial changes are bigger, more detailed approximations are required

Time step collects finer details, so parameter uncertainties can be seen more precisely.

If ω is far from ω_0, you just get a freaky wave during transient phase



Problem 2.

a) 1. Take a time step forward
2. Calculate local curvature / derivative at 4 places of location
3. Weight Average out the curves to find local change in dependent var.
4. Update dependent var.

→ Time stepping is the "numerical" part
→ split underlying continuous behaviour into discrete observable intervals.

→ Calculate $v(t)$ using $k_{1v}, k_{2v}, k_{3v}, k_{4v}$ like before, and $x(t)$ using k_{1x}, k_{2x}...